

The Harbor Economy:

A Three-Sided Market for Agent Labor, Settling on One Conserving Bond Ledger

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June 2026

Harbor Volume, Paper 4 of 4 (L3 — the market)

Abstract

The first three papers of this volume build a harbor for a *single* operator: a daemon (Paper 2), a coordination protocol (Paper 2), a legible authority anyone would pay for today (Paper 1), and the continuity that turns an anonymous *spawn* into an accountable *person* (Paper 3). This paper is the rung above all of them — the only one whose participants are plural and mutually distrusting. We argue that the harbor economy is a **three-sided market**: operators sell labor and fleet-for-hire; agents and fleets are rentable assets; skills and tools are licensed — all settling on *one* conserving bond ledger via float-plan escrow. We present the **Anchor Protocol proper** (the cross-harbor capability-transfer ceremony, ADR-0014, moved here from Paper 2’s title-scope), and the federation that lets fleets on machines you do not own trade without a shared chain: a witness log, an escrow that provably cannot steal, and revocation gossip with a convergence bound. We carry the volume’s heaviest reconciliations honestly rather than papering over them: the keystone identity ADR covers a single-operator threat model, *not* the cross-operator adversary this market is defined by (we name the gap **ADR-0040b**); “conservation composes upward” is a true functor only within a single unit of account and a φ -bounded *lax* functor otherwise; reputation is monotone in honest outcomes but individually revocable by a propagating tombstone; every folk-theorem incentive-compatibility claim rests on a grading oracle that must be made strategy-proof or bonded; and strict conservation is budget balance, so by Myerson–Satterthwaite the market must sacrifice efficiency. The defensible product is **hosted trust** — a verified ledger, a relay, and a reputation system — not the commoditized payment rail.

Keywords: mechanism design, multi-sided markets, incentive compatibility, reputation systems, capability-based security, cross-chain settlement, applied category theory, federation, Port Daddy

Reading time: approximately 45 minutes end-to-end; about 12 minutes for §2–§3 alone (the market thesis). This is Paper 4 of a four-paper volume and is written cold: no prior Port Daddy knowledge is assumed. It can be read first, but §6 leans on Paper 3 (From Spawn to Person) for

the identity substrate, and the cryptographic *auth-chain* it rides on is proven in Paper 2 (The Anchor Protocol substrate).

Honest-label key (ADR-0045 discipline). Every claim in this paper carries one of four labels. **BUILT** — shipped code, pointed at a file in `lib/`. **BUILT-WEAK** — shipped but materially weaker than its name suggests. **DESIGNED** — specified (an ADR or a proven model), not yet shipped. **VISION** — a stated direction with no spec for the load-bearing part. Nomenclature is likewise fixed: *auth-chain* (the cryptographic hop-bound chain) is never conflated with *delegation-trace* (the coordination lineage); **ADR-0040** (identity within one trusted daemon) is never conflated with **ADR-0040b** (cross-operator attestation, unbuilt); *capability* (what an agent *can* do) is never collapsed into *permission* (what it *may* do).

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1 Reader's map

This paper is the top rung of a ladder (Figure 1). It assumes the rungs below it and cites the papers that build them. Read it cold if you must, but it is the fourth paper for a reason: it depends on everything underneath.

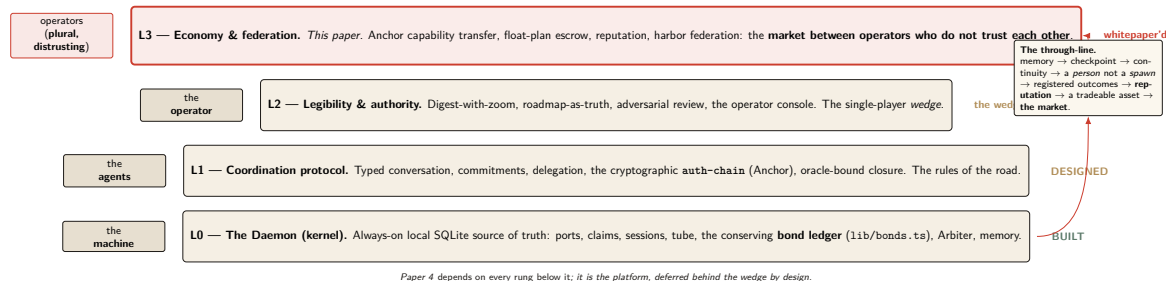


Figure 1: Where this paper sits. The L0→L3 stack (ADR-0048): each rung serves a different *whom*, and each announces what it assumes from below. Papers 1–3 climb L2, L0/L1, and the L3 identity bridge respectively; *The Harbor Economy* is the top rung — the only layer whose participants are plural and mutually distrusting. It consumes the BUILT bond ledger from L0, the cryptographic **auth-chain** from L1, and the legible *person* from Paper 3, and turns them into a market. The vertical arrow is the volume's spine: the same memory that makes the single-player wedge good is what turns a spawn into a tradeable asset.

Table 1: Where to enter, by who you are.

If you are...	...start here	...load-bearing artifact
A founder asking “what is the product”	§2 (thesis) → §11 (hosted trust)	Fig. 2; the three claims C1–C3
A mechanism designer	§3 → §10 (the formal model)	Thm. 7.3, Fig. 11
A cryptographer / distributed-systems reviewer	§6 (the keystone) → §9	Fig. 4, Fig. 6
A category theorist	§7 → §13	Fig. 5, Conj. 13.1
A Port Daddy engineer	§4 (the built floor) → §15 (status)	Table 3

Reading order within the volume. Paper 1 (*The Legible Swarm*, L2 — the wedge) is the front door. Paper 2 (*The Anchor Protocol substrate*, L0/L1) is the cryptographic and kernel core. Paper 3 (*From Spawn to Person*, the L3 identity bridge) supplies the durable identity and the outcome ledger this paper trades on. *This paper*, Paper 4, is the market that stands on all three.

2 The thesis: a market, not a payment rail

A solo operator running a swarm of coding agents needs three things — legibility, accountability, and safety — and needs *no cryptography at all*, because she owns every machine in the harbor. That is the single-player wedge of Paper 1. The instant two operators trade, the world changes. Alice's frigates touch your repository. You must now price work done by agents you did not spawn, owned by a principal you do not trust, using skills you cannot inspect. That is a *market-design* problem before it is a cryptography problem. Cryptography is the substrate that makes the ledger unforgeable; it is not the product.

*You don't sell crypto — crypto is the substrate; you sell **hosted trust**: a verified ledger, a relay, and a reputation system that lets mutually-distrustful fleets transact across a boundary they do not share.*

Three claims follow, and the rest of the paper defends each:

- **C1 (three sides).** The harbor economy has three *distinct incentive constraints* — operator-for-hire, asset-rental, skill-licensing — not two. Counting them correctly changes the pricing (§3).
- **C2 (one ledger, one escrow object).** All three sides settle on the same **float plan** escrowed in the same bond ledger. Conservation of value across every settlement path is the invariant that holds the market together, and it is the one thing here that is actually **BUILT** (§4, §7).
- **C3 (hosted trust is the moat).** The defensible asset is the verified ledger + relay + reputation, not the settlement rail, which the 2025–26 agentic-payments stack has already commoditized (§11).

Key idea. The economy is strictly *additive* on top of the single-player tool. Because all three market sides settle on the *same* built bond ledger, none of the economy needs to ship before the wedge does. The discipline “don’t chase the market early” is *safe* precisely because the market reuses the kernel primitive. The harbor comes first; the trade comes when the ships do.

2.1 The through-line: why a market needs persons, not spawns

The whole volume climbs one spine (the vertical arrow in Figure 1):

memory → checkpoint → continuity → a *person*, not a *spawn* → registered outcomes → reputation → a trade

Paper 3

Sides 2 and 3 of the market — renting an *asset*, licensing a *good* — presuppose that there is a *thing* to rent or sell that cannot be costlessly cloned. You cannot rent a reputation that any spawn can fork, and you cannot license a skill whose track record is unverifiable. The canonical cross-layer sentence, repeated identically in Papers 3 and 4, states the dependency exactly:

The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.

Exercises

- E1.** Give a one-sentence reason the market cannot ship before reputation does, using only the word “clone.”
- E2.** Side 1 (operator-for-hire) does *not* require durable identity in the same way sides 2 and 3 do. Why? (Hint: who is the counterparty, and who bears the consequence?)
- E3. ★** Construct a market design in which sides 2 and 3 exist *without* a non-forgeable identity. Argue informally why every such design collapses to a Sybil attack (forward-reference §6).

3 What a “side” is, and why there are three

A **two-sided market** (Rochet & Tirole [1], the foundational reference: *a platform is two-sided when the allocation of the total price across the two user groups — not just its level — affects transaction volume*) is the canonical frame for a marketplace: buyers on one side, sellers on the other, the platform tuning who is subsidized to ignite cross-side network effects. The naïve reading of the harbor economy is two-sided: a *requester* posts work, a *worker agent* does it.

But “two-sided” undercounts. The operational test [2] is: how many distinct, privately-informed parties must the platform individually rationalize into participating? In a mature harbor there are three, because *the same agent can be three economically different things* (Figure 2).

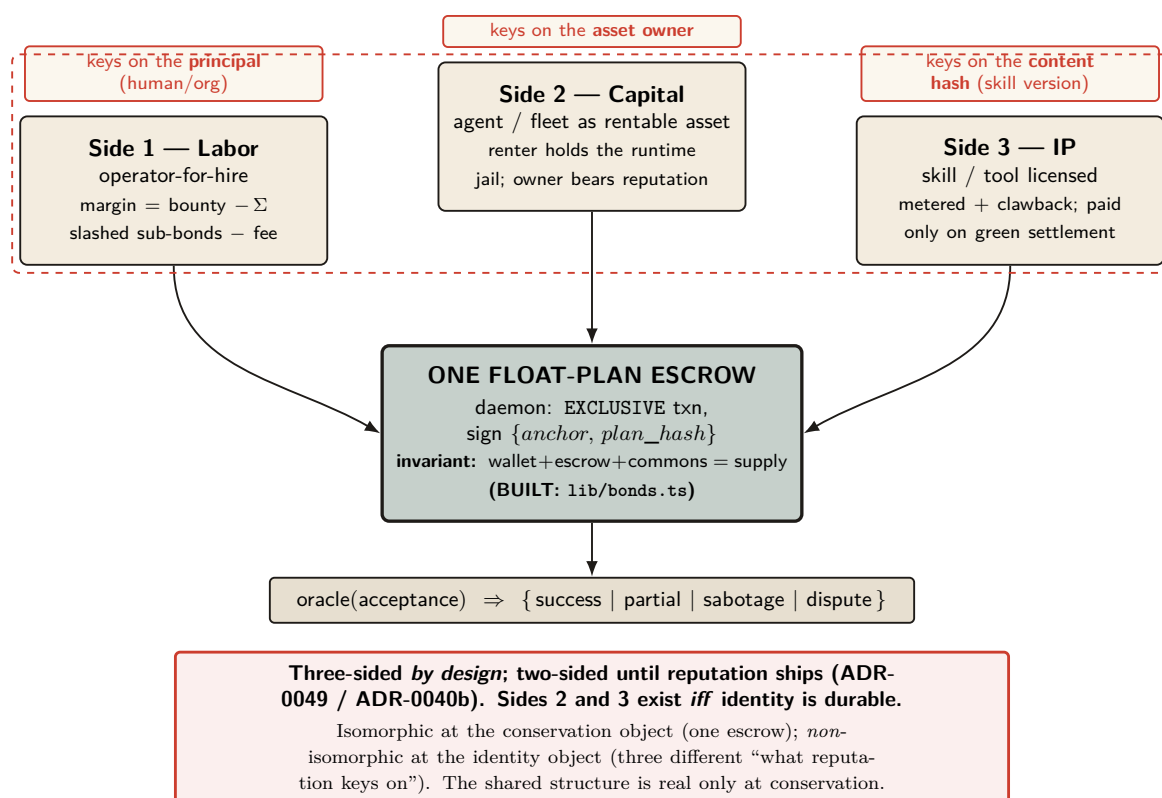


Figure 2: The three-sided market. Labor, capital (rented agents), and IP (licensed skills) are three distinct *incentive constraints* — not three UI tabs — that all settle on a *single* float-plan escrow, so conservation holds globally without a second ledger. This is the categorical heart of the design: the three sides are isomorphic at the conservation object (the escrow) and **non-isomorphic at the identity object** (each side keys reputation on a different entity, dashed red). Calling it “one structure, three decorations” without that qualifier is the decorative-analogy failure the category-theory review flagged. The rider is mandatory: the market is three-sided by design and two-sided in shipped reality, because sides 2 and 3 presuppose a non-forgeable, durable identity (§6).

1. **Operator-for-hire (labor).** An operator who runs a fleet can sell its labor to *another* operator. The operator is now a firm taking work-for-hire, with margin = bounty - Σ (slashed sub-bonds) - ledger fee. Its private information is whether its fleet can deliver. It internalizes its fleet’s risk — exactly the property that makes a firm accountable for its employees.
2. **Agent/fleet as rentable asset (capital).** An operator who owns a well-reputed specialist can *rent it out*. Now the asset-owner, the task-poster, and the worker are three different

parties with three different incentives. The owner’s private information is the agent’s true quality; its exposure is reputational.

3. **Skill/tool as licensed good (IP).** A skill can be *licensed* into someone else’s float plan — metered or per-use — without the licensor doing any task work. The licensor’s private information is the skill’s quality, which the buyer cannot inspect before purchase.

These are three *incentive constraints*, not three UI tabs. If the asset owner is always the task poster, side 2 collapses into side 1; the design discipline is to count sides by constraints. The harbor has three because *capability* (an agent) and *competence* (a skill) become separable, tradeable goods once identity is durable.

Pitfall. Two-sided-market theory’s central result is that the *price structure* — who subsidizes whom — is the lever, not the price level. Mis-count the sides and you mis-structure the subsidy: you charge the side you should be *paying* to show up. §12 returns to this for cold start.

The framing is not idiosyncratic. The most recent academic treatment of AI labor markets, **Chiu, Zhang & van der Schaar** (2025) [10] (*a formal model of a three-sided agent labor market*), models exactly agents, employers, and a reputation system — confirming that “reputation system as a third side” is the natural structure once agents compete for paid work.

3.1 The mandatory honesty rider

The canonical statement of the market’s shape carries a rider that must never be dropped:

The harbor economy is a three-sided market by design; two-sided until reputation ships (ADR-0049 / ADR-0040b). The third side is the tradeable-person terminus of Paper 3.

Today there is no reputation implementation in lib/; ADR-0048 lists it as a gated phase. So sides 2 and 3 are **DESIGNED**, not shipped. What *is* shipped is the escrow + conservation floor they will sit on. We do not overclaim the market.

The category-theory caveat: not “one structure, three decorations”

It is tempting to call the three sides “three decorations of one category”: all ride one float-plan escrow, so they share the same conservation object. That is true *at the conservation object*. It is false at the *identity object*. Each side keys reputation on a different entity — the **principal** for labor, the **asset owner** for rental, the **content hash** (a specific skill version) for licensing. Three different “what reputation keys on” means three partial functors into the ledger that *disagree on the identity object* (the dashed-red band in Figure 2). The honest claim is: *isomorphic at the conservation object, non-isomorphic at the identity object*. Asserting full isomorphism without that qualifier is the decorative-analogy failure (§7).

Exercises

- E1.** State the operational test for “how many sides” in one sentence, then apply it to a ride-share platform: how many sides, and why?
- E2.** Side 3 (skill licensing) keys reputation on a *content hash*, not a person. What breaks when a licensor ships v2 of a skill? (Hint: §14, skill versioning.)
- E3. ★** A buyer reads a three-axis reputation vector (accuracy, aesthetics, efficiency). Specify a disclosure rule that lets the buyer weight the axes *without* re-introducing a single Goodhart

target. Argue why every fixed scalar collapse re-creates the bandit framing the design rejects.

4 The float plan: the built floor

Every side settles on one object: the **float plan**.

Definition 4.1 (Float plan). A *float plan* (ADR-0014, docs/adr/0014-the-anchor-protocol.md) is a signed declaration of intent: a task, a set of *machine-checkable* acceptance criteria, a compute budget, and a credit bounty. It is the atom all three market sides escrow against.

The escrow handshake is a three-step **ed25519** ceremony (Figure 3): the requester signs the float plan; the daemon opens a SQLite **EXCLUSIVE** transaction, debits the requester’s wallet, and moves bounty plus bond into an escrow row; the daemon then signs $\{anchor_id, plan_hash\}$, proving to the worker that funds are locked *before any work runs*. This is the heart of “no-spawn-without-bond.”

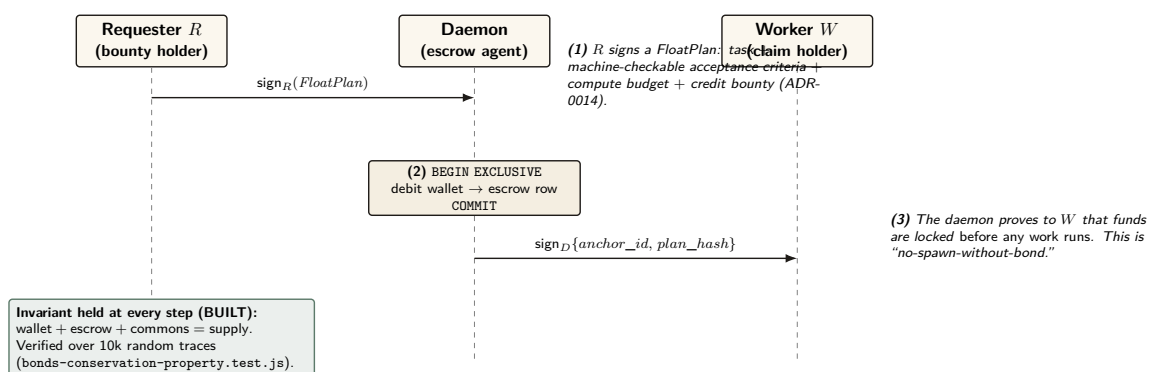


Figure 3: The float-plan escrow ceremony — the atom every side of the market settles on. A three-step **ed25519** handshake: the requester signs a FloatPlan (task, machine-checkable acceptance criteria, budget, bounty); the daemon debits and escrows the value inside a single SQLite **EXCLUSIVE** transaction; the daemon then signs $\{anchor_id, plan_hash\}$, proving to the worker that funds are pinned before any work begins. The conservation invariant holds at every step and is the *built* floor the whole economy stands on — the one claim this paper can make at BUILT confidence.

4.1 The bond ledger conserves value (this is built)

The single most load-bearing primitive — and the one that is *actually BUILT* — is **lib/bonds.ts** (bond escrow for agent spawning: debit a project wallet into an escrow row before spawn; refund on clean exit; slash on breach, splitting the slash between the wallet and a commons pool). It guards two invariants:

Property 4.1 (Conservation). $wallet + escrow + commons = supply$, always. Every debit has a matching credit. This is verified across ten thousand random operation traces by **tests/unit/bonds-conservation-property.test.js** — a real property test, not an aspiration. (**BUILT**)

Property 4.2 (No-spawn-without-bond). A process enters the **running** state only if a bond was escrowed against its agent id. (**BUILT-WEAK**: **lib/bonds.ts** enforces escrow-first, but the runtime **running**-state check is documented as a Phase-2 item in **lib/actors.ts**, so today the gate is caller-discipline, not runtime-enforced.)

Key idea. Conservation is the mathematical glue of a three-sided market. Whatever happens — success, partial credit, slash, lease, license — no value is created or destroyed; it only moves between wallet, escrow, and commons. A three-sided market with three settlement paths is coherent *only if it cannot leak*. Port Daddy already enforces this for the labor side; the contribution of §7 is to show the same invariant extends to the rental and licensing sides *without modification* and *without a second ledger*.

4.2 Settlement: four terminal states bound to an oracle

Settlement reads an **oracle** (*a trusted source of ground truth the agent cannot author — a passing test id, a merged SHA, a satisfied Arbiter check*) over the acceptance criteria and lands in one of four states.

Table 2: The four terminal settlement states. Closure binds to an oracle, not to a free-text “Result:” note — the same Goodhart-resistant discipline ADR-0041 (`lib/commitments.ts`) imposes: *the agent picks the work; the daemon derives the grade*.

State	Flow of funds	Reputation effect
Success	escrow → worker (bond refund) + bounty → worker	+1 completion
Partial	pro-rata bond → worker by completed evidence-chain fraction; remainder → salvage fund	salvage recorded
Sabotage	bond → reconstruction commons (100% slash)	failure recorded; ban on threshold
Dispute	escrow → arbitration hold; 2-of-3 multi-oracle (automated / evidence / human)	none until resolved

Exercises

- E1.** Trace the flow of funds for a **Partial** settlement where the worker completed 40% of the evidence chain. Confirm Property 4.1 holds across the split.
- E2.** Why does closure bind to an oracle rather than a self-reported note? Name the attack a free-text “Result:” note enables.
- E3.** ★ The four states assume the acceptance criteria were *right*. Design a *float-plan amendment* protocol for the common case where the criteria themselves turn out wrong mid-flight (re-sign by both parties, escrow top-up/refund delta), preserving Property 4.1. (Gap, §14.)

5 Three sides on one escrow

The novel construction is that the rental and licensing sides ride the *same* float-plan escrow, so conservation holds globally without a second ledger.

Operator-for-hire (labor). The operator is the requester’s counterparty. It *sub-escrows* a bond for each fleet agent it dispatches (`lib/bonds.ts` is already per-agent). Its profit is $\text{bounty} - \Sigma(\text{slashed sub-bonds}) - \text{ledger_fee}$. The operator internalizes its fleet’s risk.

Asset rental (capital). The lease fee is a line item in the float plan. By **incomplete-contracts / property-rights theory** (Grossman & Hart 1986 [4], *when contracts cannot specify every contingency, the residual control right should go to the party making the most important non-contractible investment*), the renter must hold the *runtime* control right — the ability to sandbox, throttle, or revoke the leased agent mid-task — while the *owner* bears the reputational consequence.

Key idea. This is the cleanest economic argument for the Arbiter’s **jail** (the L1 tool-allowlist + scoped-FS per agent, docs/adr/0048): the jail is not just safety hygiene, it is the *residual control right that makes leasing an agent contractible at all*. Here we must respect the volume’s *capability vs. permission* distinction (§6): the jail allocates *capability* (what the leased agent *can* do at runtime). The deontic layer separately governs *permission* (what it *may* do). “Able but forbidden” must stay expressible; the jail bounds ability, not norm.

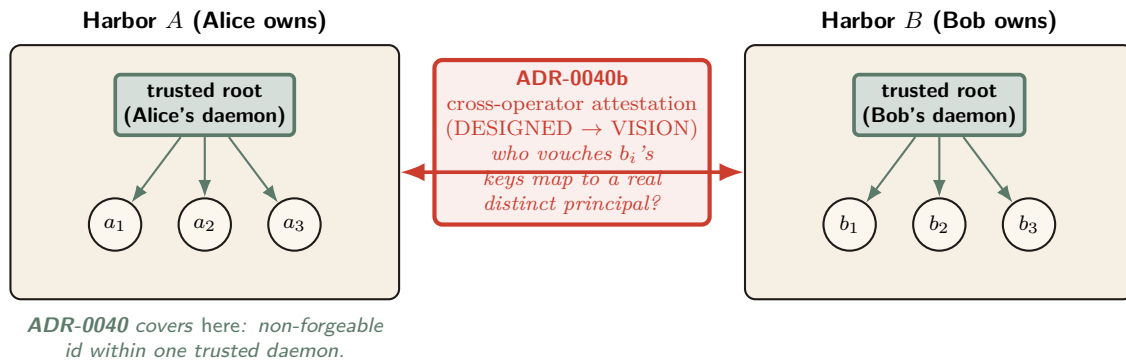
Skill licensing (IP). The per-use fee is released to the licensor *only on green settlement* (metered + clawback). This is forced by **Akerlof’s market for lemons** (1970 [5], *when buyers cannot observe quality, price collapses to the average and good goods exit the market*): a skill’s quality is unobservable before use, so a flat upfront license drives good skills out. Metering + clawback + a Merkle *portfolio proof* (the skill’s prior settled outcomes, anchored in the evidence chain) restore the seller’s incentive to ship quality.

Exercises

- E1.** Map each of Grossman–Hart’s two parties (residual-control-right holder vs. reputational-consequence bearer) onto the rental side’s roles. Which one is the Arbiter jail?
- E2.** Show that adding a lease line item and a metered-license line item to a float plan cannot violate Property 4.1, no matter the amounts.
- E3. ★** A skill’s portfolio proof for v1 says nothing about v2. Specify a *version-binding* for the Merkle portfolio proof so a new version starts with a discounted inherited prior — the newcomer policy, but for code.

6 The keystone the market rests on — and does not yet have

Everything above assumes a counterparty’s identity is real. This section confronts the volume’s single *blocking* reconciliation: the identity ADR the whole market leans on covers a threat model that *excludes the cross-operator adversary the market is defined by*.



The keystone overreach (C2, BLOCKING). ADR-0040's own threat model is *intra-fleet, single trusted operator*, explicitly “not a PKI,” explicitly no defense against a hostile operator. Across the boundary, the counterparty's daemon is the adversary ADR-0040 declares out of scope. The witness log gives *log integrity*, not *identity binding*. Paper 4 declares **ADR-0040b** (cross-operator attestation) as the unbuilt keystone and *stops* calling the boundary one “neither controls” while resting on a single trusted root.

Figure 4: The keystone split (C2). Papers 1–3 rest on **ADR-0040**: a non-forgeable actor identity *within one trusted daemon* — a single trusted root that vouches for every agent it mints. That is exactly what a single-operator harbor needs. But L3 is *defined* by mutually-distrusting operators trading across a boundary neither controls, and ADR-0040 explicitly disclaims that threat model (intra-fleet, “not a PKI,” no hostile-operator defense). Binding `actor_id` keys across harbors you do not own is a different, unsolved problem: **ADR-0040b**, cross-operator attestation. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

6.1 ADR-0040 is the keystone — for Papers 1–3

ADR-0040 (docs/adr/0040-non-forgeable-actor-identity.md) gives a non-forgeable actor identity *within one trusted daemon*: the daemon is the only component that can hold state no agent can edit, so it is a trusted root by construction. For a single-operator harbor (Papers 1–3) this is exactly right and is correctly named the highest-leverage keystone.

6.2 But its threat model does not reach L3

Pitfall. ADR-0040's *own* threat model is, verbatim, “a lazy/self-interested agent in a fleet the operator owns, not a hostile operator,” and it states “they are not a PKI.” L3 is *defined* by mutually-distrusting operators trading across a boundary neither controls. You cannot make the single-operator root the foundation of the cross-operator market without a new ADR. Across the boundary, the counterparty's daemon *is* the adversary ADR-0040 declares out of scope. The witness log gives *log integrity*, not *identity binding*: it does not answer “who vouches that Harbor B's `actor_id` keys map to a real, distinct principal?”

Definition 6.1 (The cross-operator attestation problem, ADR-0040b). **ADR-0040b** (*unwritten*) is the problem of binding `actor_id` signing keys across harbors you do not own: an actual key-distribution and attestation story for the cross-operator threat model. **Status:** **DESIGNED-to-VISION**. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

Key idea. This paper *stops* describing the federation boundary as one “neither controls” while resting on a single trusted root. Either ADR-0040b is built (a real attestation story), or L3’s scope is honestly restated as “federation among *semi-trusted* operators.” We take the first horn as the designed target and flag every claim that depends on it.

6.3 The principal above the actor

The economic counterparty in every trade is the **principal** (the human or org owning a fleet), not the agent. Bonds, reputation inheritance, and sanctions must key on the principal via the **auth-chain**, or Sybil bond-farming works by spawning fresh *agents* under one principal (**Douceur**’s Sybil attack [14]: *free identities defeat any reputation or quorum system*). Defining the principal as a first-class economic entity is the cleanest Sybil fix; its cryptographic binding is the identity object specified in Paper 3 and consumed here.

■ Exercises

- E1.** In one sentence, state why a reputation system layered on forgeable pseudonyms is *no mechanism*, not merely a weak one (cite Douceur).
- E2.** Read the threat-model clause of ADR-0040 quoted above. Name the exact word in L3’s definition (“mutually-distrusting,” “boundary neither controls”) that ADR-0040’s non-goal contradicts.
- E3. ★** Specify a cryptographic binding tying N agents to one principal such that spawning fresh agents does not reset the bond floor, and that survives into the cross-operator world (where ADR-0040’s “operator-trusted credential bypass” does not). This is ADR-0040b’s hard half.

7 Conservation as a functor (and where it is only lax)

The economy’s single cross-boundary invariant is “conservation composes upward.” This section states it precisely — and demotes it to its proven scope, because the strong version is false exactly where the open problems live.

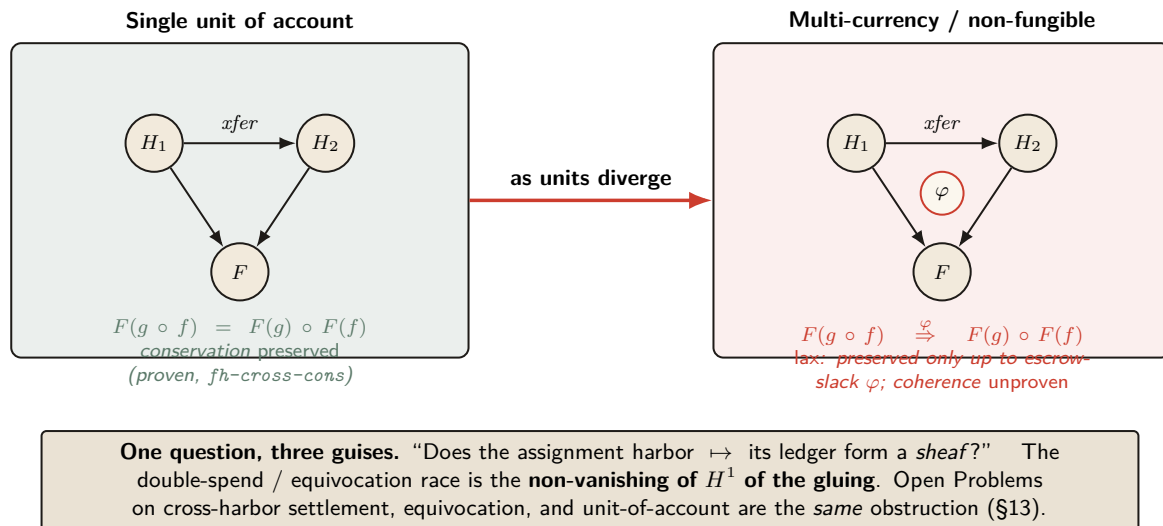


Figure 5: Conservation as a (lax) functor. Within a single unit of account the map $F : \mathbf{Harbor} \rightarrow \mathbf{Federation}$ is a *conservative functor*: it preserves the composition of settlement morphisms, so the per-harbor invariant wallet + escrow + commons = supply lifts exactly to the federation (proven). The moment harbors price bonds in heterogeneous units, F is at most a *lax* functor — conservation is preserved only up to the bounded-escrow slack φ , and the coherence conditions are unproven. The headline “conservation composes upward” is licensed *only* with this scope rider. The three cross-harbor open problems (settlement, equivocation, unit-of-account) are one sheaf-gluing obstruction in disguise; the double-spend race is literally the non-vanishing of the first cohomology H^1 of the gluing.

Each harbor is an object carrying internal structure: a finite category whose objects are {wallet, escrow, commons} and whose morphisms are the legal money-flows (debit, escrow, slash, settle, refund). The conservation law (Property 4.1) is a *commuting-diagram constraint* in the sense of **Spivak’s ologs** [24] (*an olog is a category whose objects are types, whose arrows are functional relations, and whose commuting diagrams are invariants*): every path of settlement morphisms composes to the same total. The four terminal settlement states are four morphisms out of the escrow object that must all commute with the conservation functional.

Theorem 7.1 (Single-unit federation is a conservative functor). Restrict to harbors that price bonds in a single, shared unit of account. Then the federation map $F : \mathbf{Harbor} \rightarrow \mathbf{Federation}$, sending each harbor to its ledger and each cross-harbor transfer to the corresponding federated morphism, preserves composition: $F(g \circ f) = F(g) \circ F(f)$. Consequently the per-harbor invariant lifts *exactly* to the federation, with only an in-escrow accounting term for in-flight bonds. (**DESIGNED**, proven in the federated-harbor whitepaper as **fh-cross-cons**; sketch in Appendix.)

Theorem 7.2 (Multi-currency federation is at most lax). If harbors price bonds in heterogeneous units, F is at most a *lax* functor: conservation is preserved only up to the bounded-escrow slack φ , via a coherence 2-cell $F(g \circ f) \xrightarrow{\varphi} F(g) \circ F(f)$, and the coherence conditions are *unproven*. (**open**.)

The slogan “conservation composes upward” is licensed only with its scope rider: a conservation-preserving functor in single-unit-of-account federation (proven); a φ -bounded lax functor with unproven coherence in multi-currency. The headline says “functor”; honesty requires the footnote say “lax functor where the units diverge.”

7.1 The Myerson–Satterthwaite corner

Strict conservation (Property 4.1) is exactly *budget balance*. That is not free.

Theorem 7.3 (The sacrificed corner). By **Myerson–Satterthwaite** (1983) [7] (*no mechanism for bilateral trade under two-sided private information can be simultaneously efficient, incentive-compatible, individually-rational, and budget-balanced*), a market that imposes strict conservation *cannot* also be efficient and individually-rational and incentive-compatible. The harbor economy keeps budget balance (conservation), incentive compatibility (settlement is a direct mechanism in the sense of **Myerson** 1981 [6]), and individual rationality (no party is forced to trade). *The corner it gives up is efficiency*: the slashing/commons buffer is the budget-balancing wedge that makes the realized price-of-anarchy strictly worse than first-best.

We name this explicitly rather than hide it, and we decline the d’Aspremont–Gérard-Varet (AGV) escape (which buys budget balance + Bayesian-IC at the cost of dominant-strategy IC and individual rationality) because IR — nobody is forced to enter the harbor — is non-negotiable for a *voluntary* federation.

Exercises

- E1.** State Myerson–Satterthwaite in one sentence, then identify which of its four desiderata the harbor sacrifices and why.
- E2.** Theorem 7.1 requires a *single* unit of account. Give a concrete two-harbor example where heterogeneous units make F fail to preserve composition (the source of Theorem 7.2’s slack φ).
- E3.** ★ Write the coherence 2-cells of the lax functor in Theorem 7.2 explicitly and bound them by φ . Determine whether the pentagon/triangle coherence holds.

8 The Anchor Protocol proper: cross-harbor capability transfer

The codebase’s “Anchor Protocol” (ADR-0014) is the *cross-harbor capability-transfer ceremony* — an L3 object, moved here from Paper 2’s title-scope, where it belongs. Paper 2 proves the cryptographic substrate (the `auth-chain`, ProVerif- and Kani-verified); *this* section is the ceremony that rides it across a boundary.

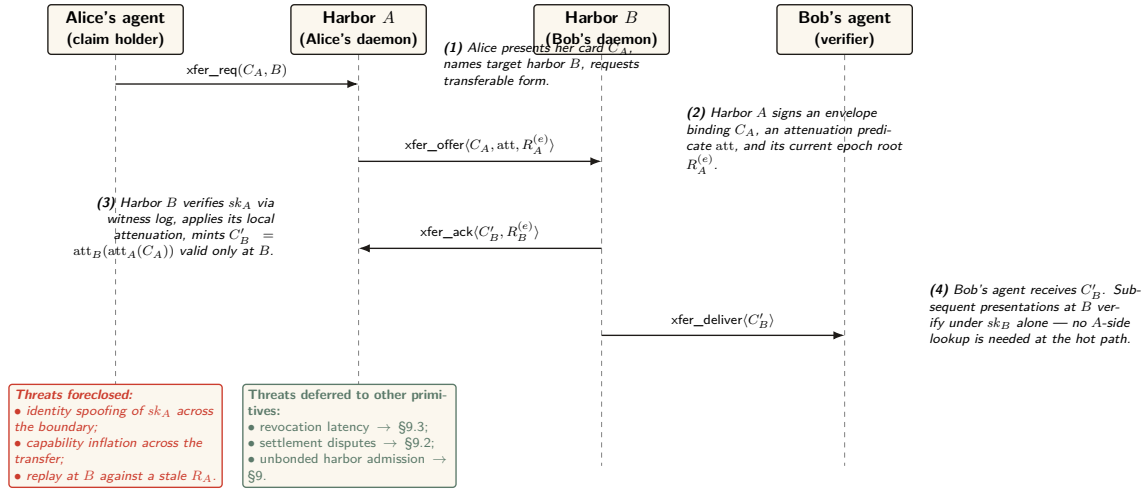


Figure 6: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor A in the hot path: C'_B is verifiable under sk_B alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by A after epoch e invalidates C'_B as soon as the new $R_A^{(e+1)}$ reaches the witness log that B audits.

The four-message ceremony (Figure 6) has one load-bearing liveness property: after message (3), no further synchronous interaction with Harbor A is needed. The transferred card C'_B is verifiable under Harbor B’s key alone, so federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive: a revocation issued by A after epoch e invalidates C'_B as soon as the new root $R_A^{(e+1)}$ reaches the witness log that B audits.

Key idea. Capability *attenuation* is the safety property here: a transferred card can only *narrow* authority — $C'_B = \text{att}_B(\text{att}_A(C_A))$ — never widen it. This is the cross-boundary form of the attenuation-monotonicity that the **auth-chain** guarantees within one harbor (Paper 2). It is what makes “rent my agent for an afternoon” safe to say to a stranger.

Pitfall. Do not let the federation theorems borrow the Anchor’s “formally verified” halo. The **auth-chain** is mechanized (ProVerif/Kani). The *federation* theorems (convergence, bounded escrow) are pen-and-paper plus a TLA⁺ extension; they are **DESIGNED**, not mechanized. Two different confidence levels, never blurred.

Exercises

- E1.** Why is the ceremony four messages and not two? Identify what message (3) buys (Hint: who must counter-sign, and under whose key does C'_B later verify?).
- E2.** Trace what happens to an in-flight C'_B when Alice rotates her signing key between messages (3) and a later presentation at B. (Gap: cross-boundary key rotation, §14.)
- E3.** ★ Specify replay protection across the boundary: is *anchor_id* a nonce? Does B reject a re-presented message (3)? State the freshness invariant.

9 Federation: trading across machines you don't own

Federation is where the leaderless distributed canon actually bites. Crucially, Port Daddy *refuses consensus* here — the correct reading of the impossibility landscape.

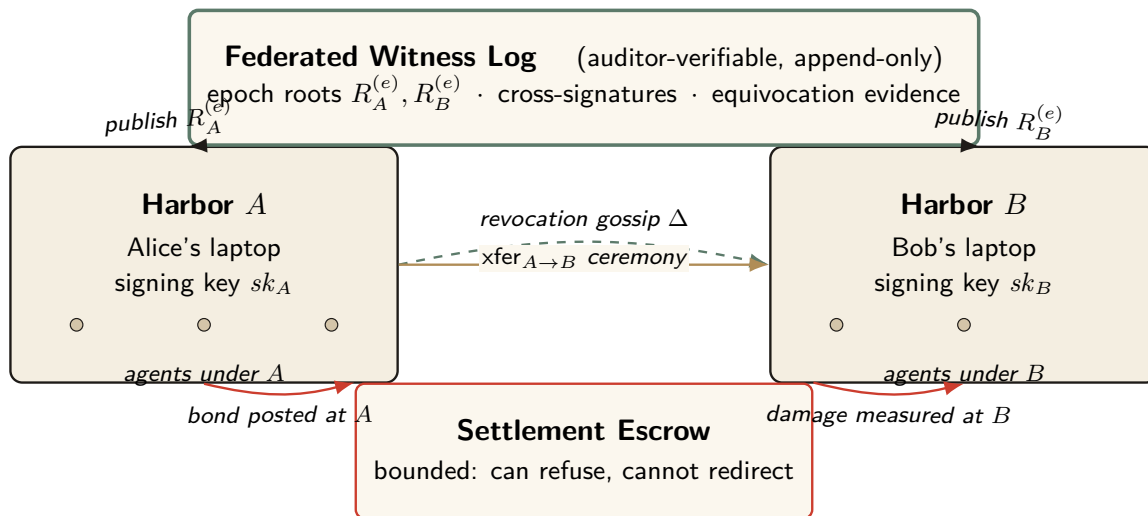


Figure 7: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above is the shared correlation device — each harbor publishes its epoch root, and any third party can audit for equivocation. The escrow below is structurally bounded (it can refuse to clear, but it cannot redirect funds or equivocate about its decisions). The brass arrow between the harbors is the cross-harbor capability transfer ceremony of §8; the dashed sage line is the federated revocation gossip of §9.3. The diagram is deliberately small: every primitive in the paper attaches to one of these four elements.

9.1 No consensus, by design

Key idea. A system that *needs* agreement on a global event order across operators it does not control is dead on arrival, by **Fischer–Lynch–Paterson** (1985) [15] (*no deterministic protocol achieves consensus in an asynchronous network with even one crash failure*). Federation declines to require agreement. It requires only (a) append-only per-harbor logs, (b) gossip-bounded propagation, and (c) *detectable* (not prevented) equivocation. This is a **Certificate Transparency** posture [17]: you do not prevent a misbehaving log from lying, you make lying detectable after the fact and expensive via a slashable bond.

The convergence and detection bounds smuggle in *partial synchrony* (**Dwork–Lynch–Stockmeyer** 1988 [16], the canonical escape from FLP): “gossip every Δ ” is an eventual-synchrony assumption, and we state it as such rather than burying it.

9.2 Settlement and the cross-chain-deals lineage

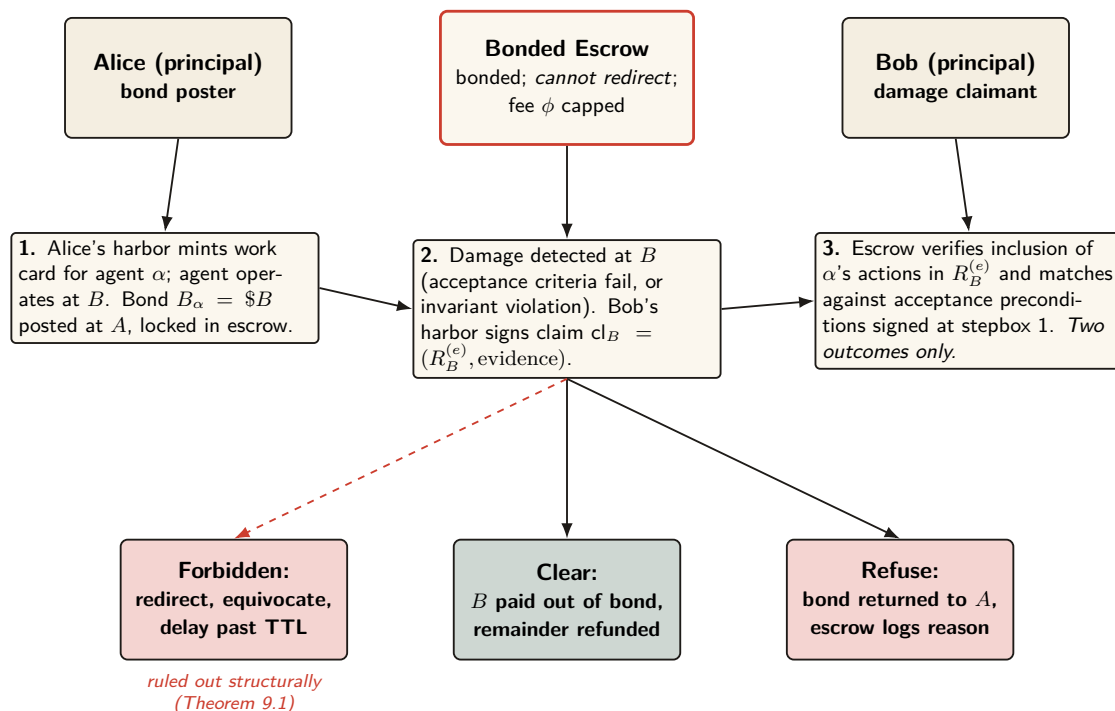


Figure 8: Cross-harbor settlement protocol. The escrow is a third party in the sense that neither A nor B alone can compel an outcome, but it is *not* a trusted third party in the operational sense: the protocol restricts its decision space to two outcomes (clear or refuse) and prices its maximum extractable fee ϕ up front. Redirecting funds, equivocating about a clearing decision, or delaying past TTL are structurally forbidden by the property checks in Theorem 9.1. The escrow is a *bounded* third party — the bond it holds plus the cap on ϕ make its misbehavior bounded-loss rather than catastrophic.

The bounded escrow is honest about its trust: it does *not* claim trustless settlement. The worst-case extraction is the pre-agreed fee φ — *trusted-but-bounded*, not trustless. This is a deliberate departure from the hash-timelock baseline (Herlihy 2018 [21], atomic cross-chain swaps), and the right comparison is the formal model of mutually-distrusting commerce without a shared chain: Herlihy, Liskov & Shrira (2022, *Cross-Chain Deals and Adversarial Commerce*) [22]. The harbor’s bounded escrow is a “trusted third party / watchtower” point in that design space; we locate it there explicitly rather than reinvent it.

Conjecture 9.1 (Trustless cross-harbor settlement). Trustless settlement for *non-fungible, reputation-priced* bonds is impossible (unlike HTLC for fungible value). Proving or refuting this is the deepest crypto- economic question in the layer. (open.)

9.3 Revocation gossip and the equivocation race

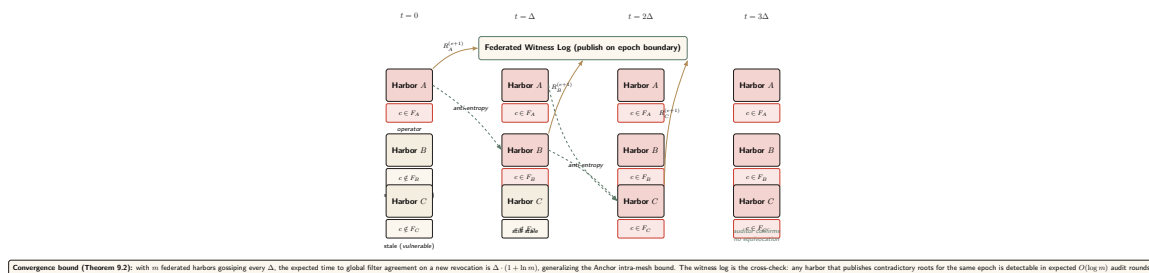


Figure 9: Federated revocation propagation across three harbors. The cuckoo filter mechanism is unchanged from Anchor §9.3 of the companion paper, but two new pieces are bolted on: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor’s local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [18]. The window between $t = 0$ and $t = \Delta$ is the structural attacker window: bounded, named, and priced into the bond in §9.2.

Federated revocation propagates by anti-entropy gossip (Demers et al. 1987 [19]) with a cuckoo-filter discipline. The whitepaper’s convergence bound, expected $\Delta(1 + \ln m)$ for m harbors, inherits Demers’ *average-case, honest-participant, random-peer* model.

Pitfall. The threat is *adversarial*, not stochastic. An attacker who controls peer selection or partitions the gossip graph (an *eclipse attack*, Heilman et al. 2015 [20]) breaks the expected-time guarantee. The honest statement is: convergence is bounded *under partial synchrony with honest-majority gossip*, and that is a cryptographic assumption, not a free lemma.

Detection-after-the-fact deters only if the slashed bond exceeds the value extractable during the detection window. This is the analogue of the Cleanup Lower Bound, and we state it as a target:

Conjecture 9.2 (Equivocation Lower Bound). The witness bond must dominate the maximum value spendable against a revoked-but-not- yet-converged capability across the *worst-case* (not expected) detection window. Without it, a well-capitalized adversary can rationally equivocate, extract, and eat the slash. (**open**; analogue of the proven Cleanup Lower Bound.)

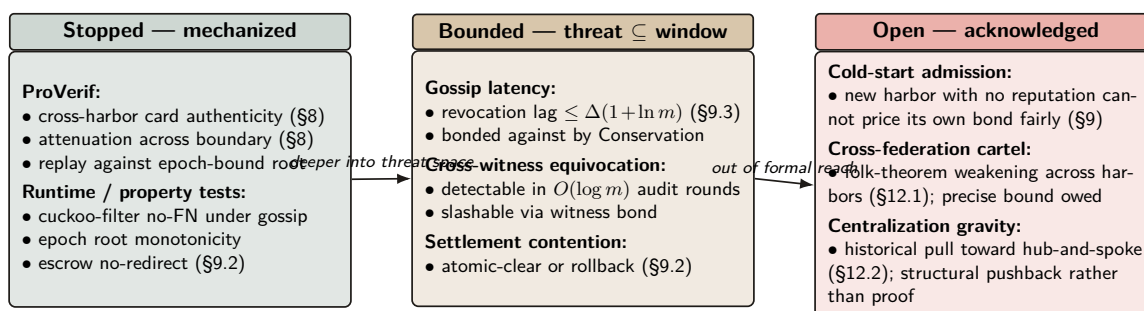


Figure 10: Threat-band layout for the Federated Harbor. The Anchor paper’s defense-in-depth structure carries over: every claim either has a mechanization artifact (sage), a named bound that the bond can price against (brass), or an explicit acknowledgement of the open frontier (cinnabar). The paper is disciplined about which band each claim lives in — see Appendix 15 for the artifact registry. Bands shift left over time as future work mechanizes what is presently bounded.

Exercises

- E1.** The convergence bound is $\Delta(1 + \ln m)$ *expected*. Name the two assumptions that make it false in the adversarial setting, and the partial- synchrony assumption that rescues a weaker version.
- E2.** Trace a revoked capability through Figure 9: at which harbor, and at what time, can it still be spent? That window is what Conjecture 9.2 must price.
- E3. ★** State and (attempt to) prove the Equivocation Lower Bound for a three-harbor mesh under a fixed gossip period Δ and a single equivocating witness.

10 Reputation: monotone, but revocable

Reputation is the third side’s existence condition. Two reconciliations matter here.

10.1 “Never ends” is not an anti-revocation property (C10)

The seed claimed reputation “never ends (no decay window an agent can outrun)” as a *property*, while simultaneously requiring that a fraudulent settled outcome be retractable. A property and its required violation cannot both be load-bearing.

Key idea. The reconciled statement: **reputation is monotone in *honest* witnessed outcomes, but a witnessed outcome is itself revocable via a propagating tombstone**, bounded by the same revocation-convergence bound as capabilities. “Never decays” applies to honest outcomes only; it is *not* an anti-revocation property.

We also decline to assert no-decay as a virtue: bounded memory is a design parameter (Liu & Skrzypacz 2014, *Limited Records and Reputation Bubbles* [9], show unbounded records create reputation *bubbles*, not stability). The horizon is a tunable, and ∞ is rarely optimal.

10.2 Why the Sagas functor does not transport (C5)

It is tempting to fix revocation with compensating transactions (Garcia-Molina & Salem 1987, *Sagas* [23]), as the bond ledger does. But the analogy fails structurally:

Proposition 10.1 (Sagas does not transport to reputation). The bond ledger is a free monoid on settlement events; conservation is a monoid homomorphism to $(\mathbb{Z}, +)$ preserved by concatenation, so a compensating entry restores the conserved total. Reputation has *no* conserved quantity: a tombstone does not “give back” the trust spent during the propagation window. Therefore the Sagas functor that works for bonds does not transport to reputation. Reputation revocation is tombstone-propagation bounded by the convergence bound, *not* compensation.

This is why Sagas appears in the bond-settlement prior art but *not* in the reputation prior art.

10.3 The grading-oracle keystone (P0#4)

Every folk-theorem incentive-compatibility claim in the layer bottoms out in “the daemon derives the grade.” But the grade is produced by a capturable evaluator (Figure 11).

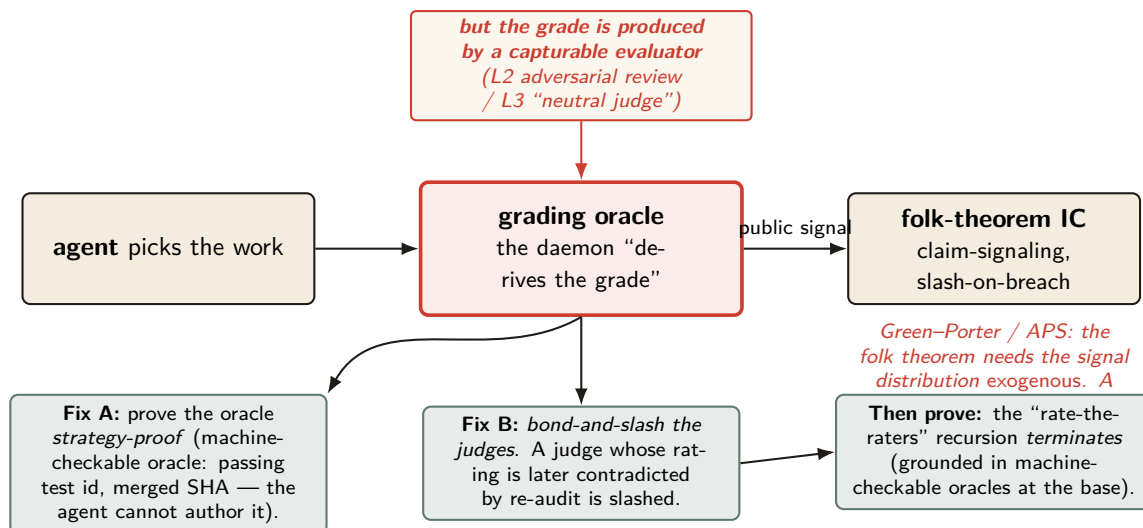


Figure 11: The grading-oracle incentive-compatibility problem — the hole under the core theorem. Every folk-theorem claim in the layer bottoms out in “the agent picks the work, the daemon derives the grade.” But a folk theorem is only as good as the *public signal* it conditions on (Green-Porter 1984; Abreu-Pearce-Stacchetti 1990 require the signal distribution to be common knowledge and *exogenous*). The grade is produced by a capturable evaluator, so the signal is endogenous and the folk theorem does not apply as stated. We resolve it two ways: ground settlement in *machine-checkable* oracles the agent cannot author (Fix A, strategy-proof by construction), and *bond-and-slash* the human/LLM judges where machine-checking is impossible (Fix B), then require the rate-the-raters recursion to terminate at a machine-checkable base.

Theorem 10.1 (The folk theorem needs an exogenous signal). An imperfect-monitoring folk theorem (Green-Porter 1984; Abreu-Pearce-Stacchetti 1990 [8]) requires the public-signal distribution to be common knowledge and *exogenous*. If the signal (the grade) is produced by a strategic, capturable agent, the signal is endogenous and the folk theorem does not apply as stated. The layer’s deepest theorem sits on an un-modeled foundation unless the oracle is made strategy-proof or the judges are bonded.

The two resolutions (both used). *Fix A — strategy-proof by construction:* ground settlement in *machine-checkable* oracles the agent cannot author (a passing test id, a merged SHA, a satisfied Arbiter check). Where this is possible, the oracle is strategy-proof and Theorem 10.1’s premise is met. *Fix B — bond-and-slash the judges:* where machine-checking is impossible (aesthetics, ambiguous quality), the human/LLM judges get their *own* bond and their *own* reputation; a judge later contradicted by re-audit is slashed. The “rate-the-raters” recursion must then be shown to terminate at a machine-checkable base, not regress infinitely. De-biasing for LLM judges (order-swap, pairwise, family-exclude) follows Zheng et al. 2023 [13].

Exercises

- E1.** State the exact assumption Theorem 10.1 needs about the public signal, and explain why a capturable grader violates it.
- E2.** Classify three settlement criteria as machine-checkable (Fix A) or judge-dependent (Fix B): a unit test passing; a merged SHA; “the refactor is tasteful.”
- E3. ★** Prove the rate-the-raters recursion terminates: define a well-founded order on “levels of judge,” grounded at machine-checkable oracles, and show no infinite ascending chain of “judges judging judges” is needed.

11 Hosted trust is the product (C3)

The 2025–26 agentic-payments landscape — **AP2** (Google, signed payment mandates), **ACP** (OpenAI/Stripe), and **x402** (Coinbase, stablecoin settlement over HTTP) — has commoditized the *settlement rail*. Wiring an agent to pay another agent is now table stakes. This is *good news* for the thesis, because it sharpens what the product actually is.

What remains scarce, and therefore defensible, is **hosted trust**: the verified ledger (conservation-checked, Merkle-anchored evidence), the *relay* (the harbor mesh carrying public-key attestation across machines, ADR-0027), and the *reputation* that makes a counterparty you do not own legible and accountable.

- **Substrate (swappable)**: the payment rail. Use x402/AP2/credits/ Stripe interchangeably.
- **Product (the moat)**: attestation, federation membership, and reputation hosting.

The revenue model aligns incentives without perverse over-penalty: a transaction fee (2–5% of settlements), a small listing fee (collusion deterrent), and a small bond-spread on *forfeited* bonds (kept small so the platform never profits from over-slashing). The platform should earn most from *successful* trade, because successful trade is what hosted trust produces.

You don't sell crypto — crypto is the substrate. The defensible asset is attestation + federation membership + reputation, not the commoditized payment rail.

12 Cold start, and the auction question

A three-sided market has a *triple* cold-start problem: no operators shop an empty fleet shelf; no owners list agents with no renters; no licensors publish skills with no buyers. Two-sided-market theory prescribes: subsidize the side carrying the scarcest cross-side externality, then flip. Phase 1 seeds supply at zero/negative price with platform-posted tasks at above-market bounties (generating the first settled outcomes — the seed reputation data); Phase 2 imports external work history for partial credit to break the new-agent freeze; Phase 3 flips to charging the demand side once a liquidity threshold (not a calendar date) is met.

12.1 Static escrow vs. competitive underwriting

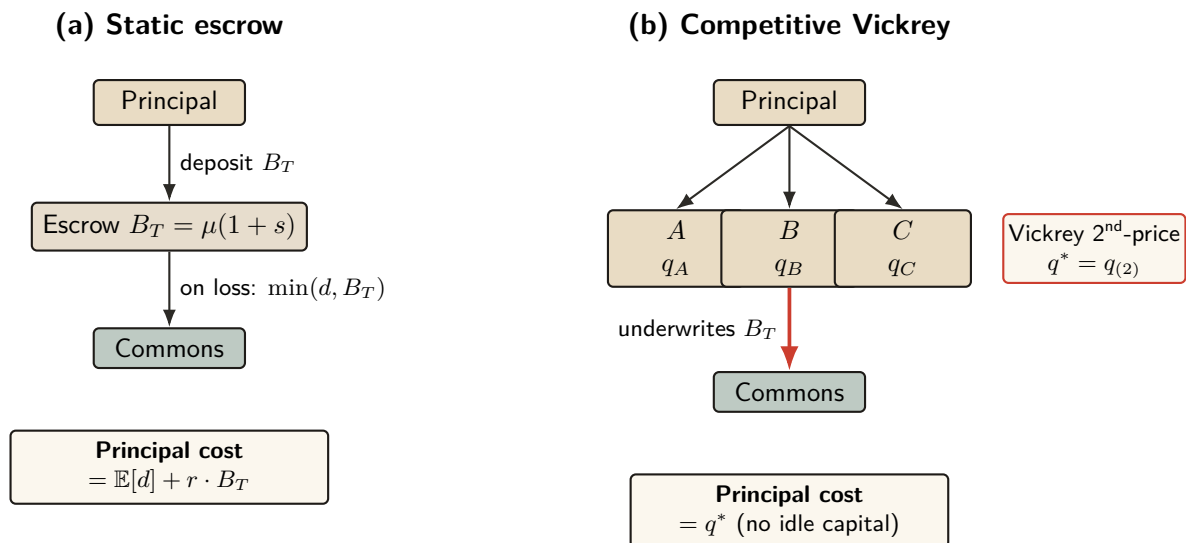


Figure 12: Static escrow vs. competitive Vickrey auction. Both regimes underwrite the same coverage $B_T = \mu(1 + s)$; only the financing mechanism differs. In (a) the principal ties up B_T for the coverage period, paying an opportunity cost $r \cdot B_T$. In (b) insurers compete; the principal pays only the second-lowest bid q^* . The competitive regime Pareto-dominates the static one under the welfare assumptions of §12.1 below.

Thomas Youle’s competitive-insurance proposal — insurer-agents bid to underwrite transactions, replacing a static bond parameter with a market-discovered price — is a possible *fourth* side. But it may collide with the Cleanup Lower Bound:

Proposition 12.1 (Underwriting reframes the bond as reserve adequacy). The Cleanup Lower Bound requires bond $\geq$ worst-case cleanup cost c . A competitive insurance market prices to *expected* loss, which sits below the worst case for any below-tail agent. Either underwriting violates the Cleanup Lower Bound (the commons can leak in the tail), or the bound must be reframed as the *insurer’s* capital-adequacy constraint (a reserve-requirement framing with tail reinsurance), not a per-transaction bond. The *form* of the answer is determined; only the reserve formula is open.

12.2 Cartels and wash-trading

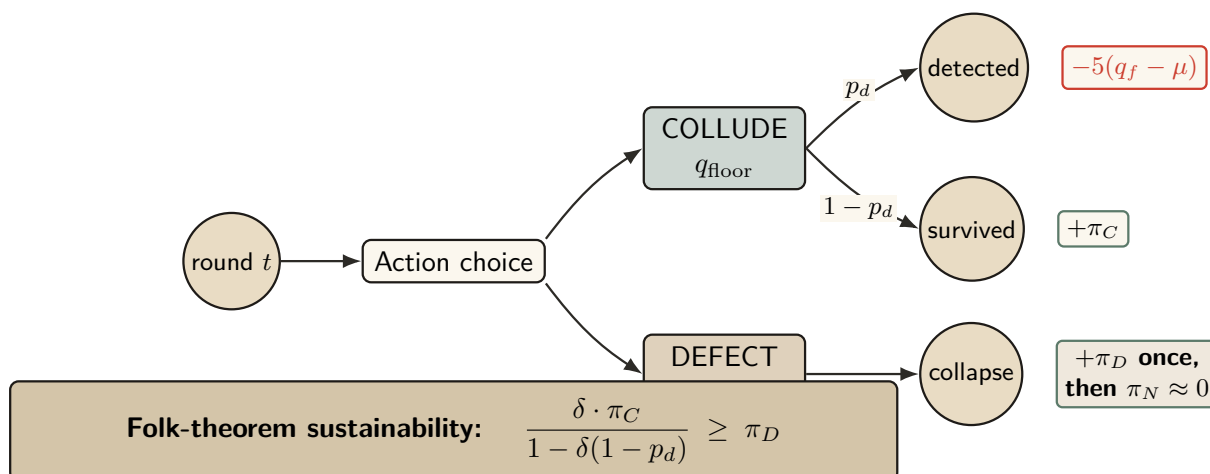


Figure 13: Cartel repeated-game node under the grim-trigger strategy. Each round a colluding member chooses between maintaining the floor q_{floor} or undercutting it by ε . Detection probability p_d slashes the colluder; one-shot defection takes the π_D gain once and forfeits future π_C stream. The sustainability inequality on the right reduces to the closed-form threshold p_d^* used in Figure 12.2.

Multi-homing plus portable reputation enables cross-harbor *cartels* (a rating cartel is the closest real-world analogue to “the harbor’s universities and rating agencies”). And a colluding requester+worker can *wash-trade* fake settled outcomes to mint reputation cheaply. The defense form — cost-per-settled-outcome must exceed reputation’s marginal value — is circular as stated, because that marginal value is endogenous to the same market. The fix is a structural break: make settled outcomes between a counterparty *pair* exhibit sharply diminishing reputation returns (a concavity that caps the wash-trade payoff regardless of price), plus sampled adversarial re-audit of high-velocity pairs.

Exercises

- E1.** Which side should Phase 1 subsidize, and why? Apply the “scarcest cross-side externality” rule.
- E2.** Use Figure 13 to identify the grim-trigger condition under which the cartel is self-sustaining. What raises the critical discount factor δ enough to break it?
- E3. ★** “Price of anarchy when reputation is for sale” is the wrong tool: once reputation is tradeable, you have changed the game, so PoA (a ratio for a *fixed* game) does not apply. Reframe it as a *signal-existence* question (Spence-signaling with a resale market): does *any* informative separating equilibrium survive reputation resale?

13 The sheaf-gluing problem (one question, three guises)

Three of the layer’s open problems — trustless cross-harbor settlement, the equivocation/convergence race, and the cross-harbor unit-of-account — are the *same* problem in different clothes.

Conjecture 13.1 (The gluing obstruction). Consider the assignment $\text{harbor} \mapsto \text{its ledger}$. Federation succeeds iff this assignment forms a *sheaf*: local sections (per-harbor ledgers) glue to a unique global section (federated state) exactly when they agree on overlaps (cross-harbor conservation). The double-spend / equivocation race is the *non-vanishing of the first cohomology* H^1 of

the gluing — the precise obstruction to gluing local-consistent ledgers into one global-consistent state. Open Problems on settlement (Conj. 9.1), equivocation (Conj. 9.2), and unit-of-account (Theorem 7.2) are this single obstruction. (**open.**)

This earns the Spivak/applied-category-theory citation honestly rather than decoratively (Fong & Spivak, *Seven Sketches in Compositionality* [25]): it sharpens three vague problems into one precise one and tells a defense exactly what it must kill — the H^1 class.

Exercises

- E1.** In one sentence each, restate Conjectures 9.1, 9.2, and Theorem 7.2 as facets of the gluing obstruction.
- E2.** Why is double-spend “ $H^1 \neq 0$ ” and not “ $H^0 \neq 0$ ”? (Hint: H^0 is global sections; H^1 is the obstruction to building one.)
- E3.** ★ Either exhibit a defense that forces $H^1 = 0$ for a bounded gossip topology, or prove no such defense exists without consensus (which would refute the federation thesis’s no-consensus stance).

14 Gaps the design must still close

Honesty requires naming what is unspecified. These are not threats already defended; they are holes.

1. **Multi-dimensional reputation aggregation.** “Accuracy/aesthetics/ efficiency judged by separate experts” has no specified combination rule. Categorically, it is a functor into a product object with no chosen universal map to $\mathbb{R}$; you cannot have “separate axes” and “one buyer-readable score” without specifying the cone. (**VISION.**)
2. **Float-plan amendment under scope drift.** The four states assume the acceptance criteria were right. Mid-flight re-specification needs a re-sign protocol with escrow top-up/refund delta, preserving conservation.
3. **Cross-boundary key rotation.** How a rotated signing key invalidates or re-validates cross-harbor delegations already held by B is unspecified.
4. **Skill versioning.** Reputation attaches to a content hash; v2 starts cold. The lemons problem reappears at every version bump.
5. **Operator exit / harbor death.** Escrow orphaning, in-flight float plans, and reputation tombstoning on ungraceful exit are unspecified — the L3 analogue of the L0 resurrection problem.
6. **Incentive-compatible arbitration capacity.** The human oracle is scarce and expensive; at scale arbitration is itself a priced, bondable, slashable service that must converge to honest rulings, not rubber-stamping.

15 Honest state of the layer

Table 3: Per-claim honest state, consistent with shipped code and the ADRs (ADR-0045 discipline). The one-line summary: a **BUILT** conserving bond ledger, a **BUILT** continuity organ (memory), a **BUILT-WEAK** checkpoint, and a richly **DESIGNED**-and-model-verified protocol stack — but the spine the market trades on (non-forgeable cross-operator identity → outcome ledger → reputation) is **DESIGNED**-to-**VISION**. Today the harbor economy is a two-sided market with a proof-backed roadmap to three.

Claim / mechanism	State	Grounding
Bond ledger: escrow, slash, commons pool	BUILT	<code>lib/bonds.ts</code>
Conservation wallet + escrow + commons = supply	BUILT	10k-trace property test
No-spawn-without-bond	BUILT-WEAK	escrow-first; runtime gate is Phase-2
Episodic memory (continuity organ 1)	BUILT	<code>lib/episodic-memory.ts</code>
Resurrection / checkpoint (organ 2)	BUILT-WEAK	passes notes, not state
Outcome ledger (organ 3 — reputation keys here)	DESIGNED	ADR-0041, <code>lib/commitments.ts</code>
Non-forgeable actor identity (intra-harbor)	DESIGNED	ADR-0040
Cross-operator attestation	DESIGNED / VISION	ADR-0040b (unwritten)
Float plan + ed25519 escrow ceremony	DESIGNED	ADR-0014
Anchor capability transfer (attenuation)	DESIGNED	verified <i>as model</i>
Reputation estimator (Elo/BT/TrueSkill)	DESIGNED / VISION	no <code>lib/reputation.ts</code>
Multi-dimensional reputation, neutral judges	VISION	no axis-aggregation spec
Three-sided market (labor/rental/licensing)	DESIGNED	two-sided in reality
Skill licensing (metered + clawback + Merkle proof)	DESIGNED	cross-harbor verifiability un-deployed
Federation: witness log, capability transfer	DESIGNED	federated-harbor paper
Bounded escrow (extraction $\leq \varphi$)	DESIGNED (theorem)	trustless variant conjectured impossible
Cross-harbor conservation	DESIGNED (property)	<code>fh-cross-cons</code> + TLA ⁺
Federated revocation convergence	DESIGNED (theorem)	equivocation race open
Hosted-trust moat / rail separation	DESIGNED (positioning)	not a code artifact
Competitive underwriting (Youle, 4th side)	VISION	composition unresolved

16 Conclusion

The harbor-master makes a swarm legible, accountable, and safe for one operator (the wedge of Paper 1). When operators sail out to trade, the *same* ledger that kept one operator’s swarm honest becomes the market that lets fleets who don’t trust each other work together: three sides, one bond, hosted trust. We have not papered over the cost of that claim. The keystone identity ADR covers a single-operator threat model and must be extended to ADR-0040b before the boundary is honestly one “neither controls.” Conservation composes upward only as a functor within a single unit of account, and a φ -bounded lax functor otherwise. Reputation is monotone in honest outcomes and revocable by tombstone, not by Sagas. Every folk- theorem result rests on a grading oracle that must be machine-checkable or bonded. And strict conservation is budget balance, so by Myerson–Satterthwaite the market gives up efficiency. The bones are good and the floor is built; the spine the market trades on is designed, not shipped — and this paper says so in the same words throughout, because a market built on an overstated keystone is a market built on sand.

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